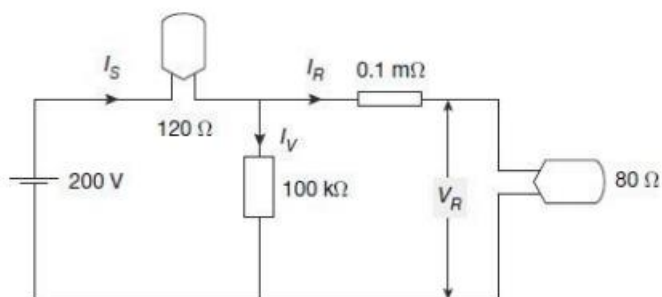
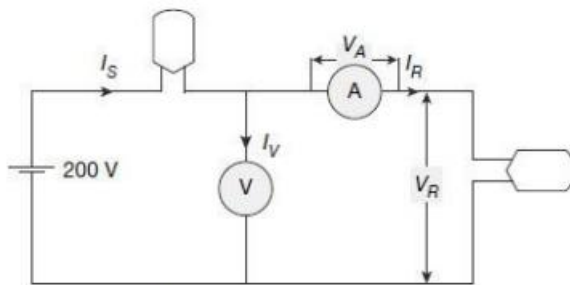
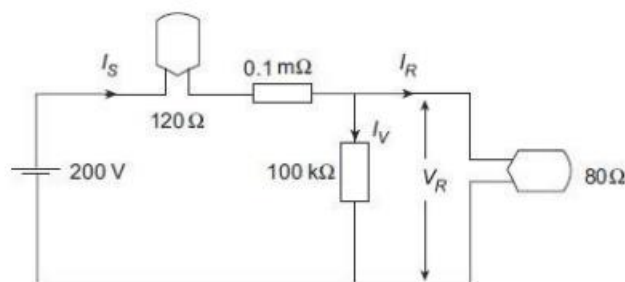
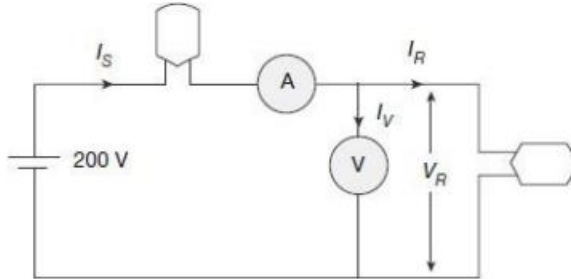
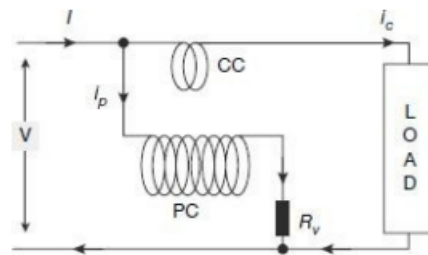


Q1: Two incandescent lamps with $80\ \Omega$ and $120\ \Omega$ resistances are connected in series with a $200\ \text{V}$ dc source. Find the errors in measurement of power in the $80\ \Omega$ lamp using a voltmeter with internal resistance of $100\ \text{k}\Omega$ and an ammeter with internal resistance of $0.1\ \text{m}\Omega$, when (a) the voltmeter is connected nearer to the lamp than the ammeter, and (b) when the ammeter is connected nearer to the lamp than the voltmeter.



Q2: An electrodynamic-type wattmeter has a current coil with a resistance of $0.1\ \Omega$ and a pressure coil with resistance of $6.5\ \text{k}\Omega$. Calculate the percentage errors while the meter is connected as (i) current coil to the load side, and (ii) pressure coil to the load side. The load is specified as (a) $12\ \text{A}$ at $250\ \text{V}$ with unity power factor, and (b) $12\ \text{A}$ at $25\ \text{V}$ with 0.4 lagging power factor.

(a) Load specified as 12 A at 250 V with unity power factor



(i) Current coil (CC) on load side

True power = $VI \cos \phi$

$$= 250 \times 12 \times 1$$

$$= 3000 \text{ W}$$

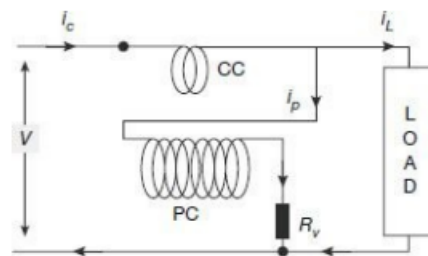
Power lost in CC = $I^2 \times r_C$ (where r_C is the resistance of CC)

$$= 12^2 \times 0.1$$

$$= 14.4 \text{ W}$$

The wattmeter will thus read total power = $3000 + 14.4 = 3014.4 \text{ W}$

Hence, error in measurement = $\frac{14.4}{3000} \times 100\% = 0.48\%$



(ii) Pressure coil (CC) on load side

$$\text{True power} = VI \cos j$$

$$= 250 \times 12 \times 1$$

$$= 3000 \text{ W}$$

$$\text{Power lost in PC} = V^2/R_p \text{ (where } R_p \text{ is the resistance of PC)}$$

$$= 250^2/6500$$

$$= 9.6 \text{ W}$$

$$\text{The wattmeter will thus read total power} = 3000 + 9.6 = 3009.6 \text{ W}$$

$$\text{Hence, error in measurement} = \frac{9.6}{3000} \times 100\% = 0.32\%$$

(b) Load specified as 12 A at 250 V with 0.4 power factor

(i) Current coil (CC) on load side

$$\text{True power} = VI \cos j$$

$$= 250 \times 12 \times 0.4$$

$$= 1200 \text{ W}$$

$$\text{Power lost in CC} = I^2 \times r_C \text{ (where } r_C \text{ is the resistance of CC)}$$

$$= 12^2 \times 0.1$$

$$= 14.4 \text{ W}$$

$$\text{The wattmeter will thus read total power} = 1200 + 14.4 = 1214.4 \text{ W}$$

$$\text{Hence, error in measurement} = \frac{14.4}{1200} \times 100\% = 1.2\%$$

(ii) Pressure coil (CC) on load side

$$\text{True power} = 1200 \text{ W}$$

$$\text{Power lost in PC} = V^2/R_p \text{ (where } R_p \text{ is the resistance of PC)}$$

$$= 250^2/6500$$

$$= 9.6 \text{ W}$$

$$\text{The wattmeter will thus read total power} = 1200 + 9.6 = 1209.6 \text{ W}$$

$$\text{Hence, error in measurement} = \frac{9.6}{1200} \times 100\% = 0.8\%$$

Q3: An electrodynameometer-type wattmeter is used for power measurement of a load at 100 V and 9 A at a power factor of 0.1 lagging. The pressure coil circuit has a resistance of 3000 Ω and inductance of 30 mH. Calculate the percentage error in wattmeter reading when the pressure coil is connected (a) on the load side, and (b) on the supply side. The current coil has a resistance of 0.1 Ω and negligible inductance. Assume 50 Hz supply frequency.

SOLUTION:

(a) Load specified as 9 A at 100 V with 0.1 power factor:

$$\begin{aligned}\text{True power} &= VI \cos \phi \\ &= 100 \times 9 \times 0.1 \\ &= 90 \text{ W}\end{aligned}$$

$$\text{Phase-angle } \phi = \cos^{-1}(0.1) = 1.471 \text{ rad}$$

Given, resistance of pressure coil circuit = 3000 Ω

and reactance of pressure coil circuit = $2\pi \times 50 \times 30 \times 10^{-3} = 9.42 \Omega$

$$\therefore \text{phase-angle of the pressure coil circuit } \alpha = \tan^{-1} \frac{9.42}{3000} = 0.00313 \text{ rad}$$

(i) Pressure coil connected on the load side

Following the expression (7.15) for actual power indication of the wattmeter in the presence of pressure coil inductance, the actual wattmeter reading is given by

$$\text{Actual wattmeter reading} = \frac{\text{True power indication}}{\cos \phi} \times \cos \alpha \cdot \cos(\phi - \alpha)$$

$$\text{Actual wattmeter reading} = \frac{90}{0.1} \times \cos(0.00313) \cdot \cos(1.471 - 0.00313) = 92.47 \text{ W}$$

$$\begin{aligned}\text{Power loss in wattmeter} &= \frac{V^2}{R_p} \\ &= 100^2 / 3000 \\ &= 3.33 \text{ W}\end{aligned}$$

$$\begin{aligned}\text{Total power indication of wattmeter, taking power loss in PC into account as well is} \\ &= 92.47 + 3.33 \\ &= 95.8 \text{ W}\end{aligned}$$

$$\therefore \text{error in measurement} = \frac{95.8 - 90}{90} \times 100\% = 6.44\%$$

(ii) Current coil connected on the load side

True power remains = 90 W

The CC gets in series with the load, and power loss in CC gets included in the total power.

$$\text{Total power} = 90 + I^2 \times r_C = 90 + 9^2 \times 0.1 = 98.1 \text{ W}$$

$$\text{Load impedance } Z = V/I = 100/9 = 11.1 \Omega$$

$$\text{Load resistance } R_L = Z \times \cos \phi = 11.1 \times 0.1 = 1.11 \Omega$$

$$\text{Load reactance } X_L = Z \times \sin \phi = 11.1 \times 0.995 = 11.05 \Omega$$

Total load resistance including CC resistance

$$R_L' = 1.11 + 0.1 = 11.21 \Omega$$

Since CC has no inductance, total load reactance including CC reactance = $X_L' = 11.05 + 0 = 11.05 \Omega$

Total load power factor including CC

$$\cos \phi' = \cos (\tan^{-1}(11.05/11.21)) = 0.999$$

$$\text{Actual wattmeter reading} = \frac{\text{Total power indication}}{\cos \phi'} \times \cos \alpha \cdot \cos(\phi' - \alpha)$$

$$\text{Actual wattmeter reading} = \frac{98.1}{0.999} \times \cos(0.00313) \cdot \cos(1.462 - 0.00313) = 100.52 \text{ W}$$

$$\therefore \text{error in measurement} = \frac{100.52 - 90}{90} \times 100\% = 11.69\%$$